FastFrac: Fast Edge Recommendation for Fractional Spectral Traces on Graphs

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***Abstract*—Budgeted edge recommendation asks which k candidate edges to add to a graph, and different analyses weight spectral scales differently. We study the positive fractional trace $F_s(S) = \mathrm{Tr}((I + \alpha L_S)^s)$, where L_S is the Laplacian after adding the edge set S, I is the identity, $\alpha > 0$ sets the regularization scale, and the exponent $0 < s < 1$ tunes the emphasis on low-eigenvalue modes. The difficulty is to compare marginal gains of this nonlinear matrix function reliably across greedy rounds and across repeated queries on one graph. FastFrac writes each marginal as a positive integral of shifted-resistance scalars; a fixed positive-weight Gauss–Jacobi rule then gives a monotone submodular surrogate with one-sided relative error. The initial scalars form a reusable index for filtered candidate pools and smaller budgets, and lazy or rank-one incremental selection attains a $(1 - 1/e - \varepsilon)$ guarantee under prescribed numerical controls, for target error $\varepsilon > 0$. Experimentally, FastFrac recovers a deterministic reference selection where a sampled Lanczos oracle takes $43.8\times$ as long and recovers 2 of 20 edges; incremental maintenance is $15.0\times$ faster than lazy selection when candidates are nearly tied; and index reuse yields 2.58–$3.57\times$ over 100-query workloads. A 4.0-million-vertex graph completes in 2.4 hours. On multiscale graphs, $s = 0.85$ gains 6.79% heat-trace reduction over log-determinant selection at diffusion time $\tau = 0.2$ but loses 10.14% at $\tau = 1$: the exponent is a horizon-dependent preference.**

***Index Terms*—Graph augmentation, spectral graph queries, submodular optimization, Gaussian quadrature, incremental computation.**

I. Introduction

Budgeted edge recommendation asks which edges should be added to a network to improve a global property. Let G_0 be a base graph with n vertices, C a pool of M admissible edges, and k the maximum number of edges to select. Unlike link prediction based primarily on local features, this task evaluates the graph-wide benefit of a selected set: adding one edge can change the value of the remaining candidates. Such problems arise in topology design for effective resistance [1], network coherence [2], and polarization reduction [3], alongside spectral graph queries [4], [5]. In graph data management, the same base network may also be queried repeatedly with different candidate subsets or budgets. This setting raises two questions: how should the benefit of adding edges be defined for different analysis goals, and how can the resulting recommendations be computed efficiently across related queries?

The choice of objective matters because an added edge can affect different parts of the spectrum. In a graph with dense communities connected by weak links, a new bridge may strongly affect slowly varying global modes, whereas an edge within a community may act more on other modes. Their effects on short-time diffusion and long-time relaxation need not agree, so the preferred edge set can depend on the analysis. Effective-resistance, network-coherence, and log-determinant criteria capture useful global properties, but each assigns its own spectral weighting. Switching between these criteria is not the same as adjusting a preference within one recommendation task. We therefore seek an objective family that lets an analyst vary the relative importance of spectral scales while keeping the edge-selection problem in a common form.

The positive fractional spectral trace provides such a family: it converts a preference over spectral scales into a graph-level score for candidate edge sets. For a selected set $S \subseteq C$, let L_0 and L_S be the graph Laplacians before and after adding S. Given a regularization parameter $\alpha > 0$ and exponent $0 < s < 1$, define $F_s(S) = \mathrm{Tr}((I + \alpha L_S)^s)$, where I is the identity matrix and Tr denotes the matrix trace. We maximize the gain $f_s(S) = F_s(S) - F_s(\varnothing)$ subject to $|S| \le k$. This choice is useful because an edge's contribution depends not only on how much it changes the spectrum, but also on which modes it affects. In particular, the first-order response of a mode with eigenvalue λ of $I + \alpha L_S$ is weighted proportionally to λ^{s-1}, together with the candidate edge's alignment with that mode. Smaller s relatively favors low-eigenvalue modes; larger s reduces that preference. Moreover, $f_s(S)/s$ approaches log-determinant gain as $s \downarrow 0$. Thus the family links a familiar criterion to continuously adjustable spectral weights. Section II-B derives this relationship, while Section V-B tests whether the resulting selections have different effects across diffusion horizons.

This modeling flexibility is useful only if recommendations can be computed efficiently at scale. Even for a fixed s, greedy selection repeatedly compares edge gains under a nonlinear matrix function. Dense spectral evaluation costs $O(kMn^3)$ over the selection process, and direct integral evaluation requires shifted linear-system solves for $O(kM)$ candidate–round pairs. Related queries on the same base graph can repeat much of this work. Positive fractional traces are known to be submodular under positive-semidefinite additions [6], so greedy approximation is possible [7], but sampling-based trace estimates do not automatically supply the valid marginal upper bounds needed for lazy selection. The computational challenge

is therefore to obtain an accurate, reusable marginal oracle that also supports certified comparisons.

FastFrac addresses these requirements through the rank-one update induced by an added edge. Integration by parts expresses its marginal as a positive integral of shifted log-determinant marginals, each depending on one resistance scalar. Adding edges decreases these scalars, making diminishing returns explicit. A fixed positive-weight Gauss–Jacobi rule preserves this submodularity with a one-sided relative-error bound, while shared Krylov computations evaluate its shifts without full-trace estimation.

The fixed shifts support an offline index of initial shifted-resistance scalars: Mq scalars and residual-based bounds for M candidates and q nodes. Filtered pools or smaller budgets reuse its rows while still paying for filtering, copying, and online selection. FASTFRAC-CELF uses lazy upper bounds; FASTFRAC-INC applies rank-one corrections when near ties make refreshes expensive. Both attain a $(1-1/e-\varepsilon)$ approximation under prescribed numerical controls, for target error $\varepsilon > 0$. Changing s or α generally requires rebuilding the index.

Our experiments address both motivating questions. For the choice of objective, on ten heterogeneous multiscale graphs $s = 0.85$ yields 6.79% more heat-trace reduction than log-determinant selection at diffusion time $\tau = 0.2$ but 10.14% less at $\tau = 1$, so the exponent acts as a horizon-dependent preference (Section V-B). For efficiency, on a 2,500-vertex small-world graph FastFrac recovers all 20 edges of a deterministic reference selection, whereas lazy selection driven by stochastic Lanczos quadrature with common random numbers (128 probes, 15 Lanczos steps) takes 43.8× as long and recovers 2. On a 400-vertex stochastic block model with nearly tied candidates, FASTFRAC-INC is 15.0× faster than FASTFRAC-CELF. Independent 100-query workloads on email-Eu-core and ca-GrQc yield 2.58× and 3.57× cumulative speedups from index reuse. FastFrac also completes a 4.0-million-vertex graph in 2.4 hours and attains mean 0.9993 OPT on small exhaustive instances.

Contributions. We make three contributions:

1) **Tunable spectral objective and certified oracle.** We characterize the spectral preference induced by s, derive a rank-one positive-integral marginal identity (Lemma 1), and construct a submodular finite-order quadrature surrogate with one-sided relative-error control (Theorem 2).
2) **Reusable index and guaranteed online algorithms.** An Mq-scalar offline index supports related queries with filtered candidate sets and smaller budgets. Residual-controlled lazy and rank-one incremental selection attain the end-to-end approximation guarantee under prescribed numerical controls (Theorem 3).
3) **Evaluation of tunability and scalability.** Numerical-oracle comparisons, near-tie workloads, repeated queries, and real/multiscale diffusion tests separate execution efficiency from objective-specific effects.

II. PRELIMINARIES

This section formalizes the edge-recommendation query, explains how the exponent controls spectral and diffusion responses, and introduces the integral tools used for efficient marginal evaluation.

A. Problem Formulation

Let $G_0 = (V, E_0, w)$ be an undirected graph, where w assigns nonnegative weights to the $m = |E_0|$ existing edges, and let $n = |V|$. The candidate pool C contains $M = |C|$ admissible non-self-loop edges, each added with unit weight. For $e = \{u, v\}$, let $e_u, e_v \in \mathbb{R}^n$ be the corresponding standard basis vectors and define $b_e = e_u - e_v$, so $\|b_e\|^2 = 2$. For a selected set $S \subseteq C$ and $\alpha > 0$, let L_0 be the weighted Laplacian of G_0, and let I denote the $n \times n$ identity matrix. Define

$$A_0 = I + \alpha L_0, \qquad A_S = I + \alpha L_S = A_0 + \alpha \sum_{e \in S} b_e b_e^\top \succeq I, \quad (1)$$

where $L_S = L_0 + \sum_{e \in S} b_e b_e^\top$ is the augmented Laplacian. Thus A_S is positive definite even if G_0 is disconnected. For $0 < s < 1$, the positive fractional trace is $F_s(S) = \mathrm{Tr}(A_S^s)$; we also write $F_{\log}(S) = \log\det A_S$ for the logarithmic criterion. To measure improvement over the base graph, use the normalized gains $f_s(S) = F_s(S) - F_s(\varnothing)$ and $f_{\log}(S) = F_{\log}(S) - F_{\log}(\varnothing)$.

Problem 1 (Fractional Spectral Edge Recommendation)**.** *Given G_0, C, $\alpha > 0$, $s \in (0,1)$, and a budget $1 \le k \le M$, find $S^* \in \arg\max_{S \subseteq C,\ |S| \le k} f_s(S)$. Write* $\mathrm{OPT} = f_s(S^*)$.

For a candidate $e \notin S$, its marginal gain is $\Delta_f(e \mid S) = f(S \cup \{e\}) - f(S)$. A function f is *submodular* if $\Delta_f(e \mid S) \ge \Delta_f(e \mid T)$ for every $S \subseteq T$ and $e \notin T$; supermodularity reverses this inequality. All approximation ratios in this paper use the normalized gain f_s. Section IV-C exploits queries that share G_0 and the spectral parameters while restricting the pool to $C' \subseteq C$ or reducing k.

B. What the Exponent Controls

To understand how s affects the relative value of candidate edges, consider the first-order response to adding an edge. Let $\lambda_1 \le \cdots \le \lambda_n$ and $\psi_1, \ldots, \psi_n$ be the eigenvalues and corresponding orthonormal eigenvectors of A_S. For a candidate $e \notin S$, let u be a scalar multiplier of its added unit weight, varying from zero to one. The trace derivative at $u = 0$ [8] gives

$$\frac{d}{du}\mathrm{Tr}((A_S + u\alpha b_e b_e^\top)^s)\Big|_{u=0} = \alpha s \sum_i (\psi_i^\top b_e)^2 \lambda_i^{s-1}. \quad (2)$$

The sensitivity depends on both the edge and the spectrum: $(\psi_i^\top b_e)^2$ measures the edge's alignment with mode i, while λ_i^{s-1} determines its spectral weighting. For two eigenvalues $\lambda_\ell < \lambda_h$, the ratio of their weighting factors is $(\lambda_h/\lambda_\ell)^{1-s}$, which decreases toward one as $s \uparrow 1$ (Fig. 1(a)). Thus, for a fixed graph and α, increasing s reduces the relative emphasis on low-eigenvalue modes. Section V-B tests how this change affects the selected edges and their diffusion behavior.

To illustrate this reweighting on the initial graph $S = \varnothing$, we measure the fraction of a candidate edge's first-order sensitivity contributed by the four lowest modes:

$$P_{\text{low}}(e,s) = \frac{\sum_{i=1}^{4} (\psi_i^\top b_e)^2 \lambda_i^{s-1}}{\sum_{i=1}^{n} (\psi_i^\top b_e)^2 \lambda_i^{s-1}}. \tag{3}$$

In (3), the eigenvalues and eigenvectors are those of A_0. Figure 1(b) compares the share across s and α for inter-cluster candidates.

The limiting cases connect this objective family to familiar criteria. Using $\lambda^s = 1 + s\log\lambda + o(s)$ for $\lambda > 0$, and extending the definition to $F_1(S) = \mathrm{Tr}(A_S)$, we obtain the following identities for unit-weight edge additions:

$$\lim_{s\downarrow 0} \frac{F_s(S) - F_s(\varnothing)}{s} = \log\det A_S - \log\det A_0,$$
$$F_1(S) - F_1(\varnothing) = 2\alpha|S|. \tag{4}$$

The log-determinant gain is recovered as $s \downarrow 0$ only after dividing the fractional-trace gain by s. At $s = 1$, all edge sets of the same cardinality have equal gains, so the objective no longer distinguishes candidates. The parameter α sets the scale of the Laplacian relative to the identity, equivalently the regularization shift α^{-1} in $L_S + \alpha^{-1} I$. Thus, α changes the spectral contrast, while s determines how the resulting eigenvalues are weighted.

The spectral response also admits a diffusion interpretation. For a fixed selected set S, define the regularized fractional generator $Q_s = A_S^s - I$. As shown below, Q_s is a weighted Laplacian and supports nonlocal fractional diffusion [9]. Let $\tau \geq 0$ denote diffusion time, distinct from the objective parameters s and α. The corresponding average heat return $p_s(\tau) = n^{-1}\,\mathrm{Tr}(e^{-\tau Q_s})$ satisfies

$$-p_s'(0) = \frac{\mathrm{Tr}(Q_s)}{n} = \frac{F_s(S) - n}{n}. \tag{5}$$

Here p_s' denotes differentiation with respect to τ. Since n is independent of S, maximizing f_s maximizes the improvement in this fractional model's initial decay rate relative to the base graph. Section V-B instead evaluates the standard-Laplacian heat trace $\mathrm{Tr}(e^{-\tau L_S})$ across diffusion times after edge selection. Thus, s controls the selection objective, whereas τ specifies an evaluation horizon; the identity above does not guarantee improvement at every horizon.

C. Integral Tools

We use integral representations of shifted inverses to evaluate noninteger matrix powers without eigendecomposition. Let A be symmetric positive definite, $t > 0$ the integration variable, and $c_s = \sin(\pi s)/\pi$. For $0 < s < 1$, the Loewner representations [10] are

$$A^s = c_s \int_0^\infty t^{s-1} A(A + tI)^{-1}\, dt, \tag{6}$$
$$A^{-s} = c_s \int_0^\infty t^{-s} (A + tI)^{-1}\, dt. \tag{7}$$

The positive-power representation underlies our marginal oracle, while the negative-power representation is used later to study

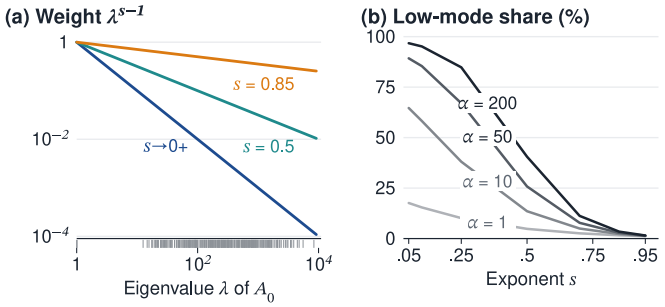


Fig. 1. How the exponent reweights the spectrum. (a) Derivative weight of (2), with ticks at the eigenvalues of A_0 on the heterogeneous reference graph of Section V-A. (b) Share P_{low} of (3) over initial inter-cluster SBM candidates, for seven exponents from 0.05 to 0.95 and $\alpha = 1, 10, 50, 200$.

inverse traces. The former also verifies the weighted-Laplacian structure of Q_s. Let 1 denote the all-ones vector. Since $L_S 1 = 0$, we have $A_S 1 = 1$ and hence $Q_s 1 = 0$. For distinct vertices $u \neq v$, rewriting $A_S(A_S + tI)^{-1} = I - t(A_S + tI)^{-1}$ gives

$$(Q_s)_{uv} = -c_s \int_0^\infty t^s (A_S + tI)^{-1}_{uv}\, dt \leq 0. \tag{8}$$

Since $A_S + tI = (1+t)I + \alpha L_S$ is a shifted graph Laplacian with an entrywise nonnegative inverse, (8) shows that Q_s has nonpositive off-diagonal entries. Together with its zero row sums, this establishes that Q_s is a weighted Laplacian.

Adding a candidate edge induces a rank-one update. For a shift $t > 0$, write $\mathscr{R}_t = (A_S + tI)^{-1}$ and define $r_e(t;S) = b_e^\top \mathscr{R}_t b_e$, the shifted resistance associated with candidate e. This is a regularized edge quadratic form, rather than the ordinary effective resistance of the unshifted graph Laplacian. The Sherman–Morrison identity [11] gives

$$(A_S + tI + \alpha b_e b_e^\top)^{-1} = \mathscr{R}_t - \frac{\alpha \mathscr{R}_t b_e b_e^\top \mathscr{R}_t}{1 + \alpha r_e(t;S)}. \tag{9}$$

These identities expose the rank-one update through shifted linear systems.

III. Theoretical Results

We derive a scalar edge-marginal identity that both avoids full-trace estimation and exposes diminishing returns. We then distinguish the positive-power objective from its negative-power counterpart.

A. Integration-by-Parts Marginal Identity

Combining (6) and (9) gives a marginal involving $b_e^\top (A_S + tI)^{-2} b_e$. Although this avoids full-trace evaluation, the squared resolvent does not directly expose diminishing returns: $X \mapsto X^{-2}$ is not operator antitone. Integration by parts instead expresses the marginal as a positive integral of logarithmic terms, each depending on a single shifted-resistance scalar.

Lemma 1. *For $e \notin S$ and $0 < s < 1$, the following absolutely convergent expressions are equal:*

$$\Delta_{f_s}(e \mid S) = \alpha c_s \int_0^\infty \frac{t^s b_e^\top (A_S + tI)^{-2} b_e}{1 + \alpha r_e(t;S)}\, dt, \tag{10}$$

$$= s c_s \int_0^\infty t^{s-1} \log(1 + \alpha r_e(t;S))\, dt. \tag{11}$$

Proof. Subtract (6) before and after adding e, apply (9), and use trace cyclicity: $\mathrm{Tr}(\mathscr{R}_t b_e b_e^\top \mathscr{R}_t) = b_e^\top \mathscr{R}_t^2 b_e =: v(t)$. This proves (10). Since $A_S \succeq I$, $0 < r_e(t;S) \le 2/(1+t)$ and $0 < v(t) \le 2/(1+t)^2$; thus the raw integrand is bounded by $2\alpha c_s t^s/(1+t)^2$ and is integrable. Moreover, $r_e'(t;S) = -v(t)$ gives

$$\frac{d}{dt} \log(1 + \alpha r_e(t;S)) = -\frac{\alpha v(t)}{1 + \alpha r_e(t;S)}.$$

Integrating by parts on a compact interval yields (11) with boundary term $-c_s t^s \log(1 + \alpha r_e(t;S))$. Its magnitude is bounded by $c_s t^s \log(1 + 2\alpha)$ near zero and $2\alpha c_s t^s/(1+t)$ at infinity; both vanish for $0 < s < 1$. The remaining integrand is $O(t^{s-1})$ at zero and $O(t^{s-2})$ at infinity. □

Equation (11) has positive weights and a logarithm increasing in $r_e(t;S)$, so the ordering of shifted resistances determines marginal comparisons.

B. Submodularity of Positive Powers and Log Determinants

Submodularity of positive fractional traces under positive-semidefinite additions was established in prior work [6]. We state the corresponding result for graph edge additions and give a direct proof using Lemma 1.

Theorem 1. *For $0 < s < 1$, f_s and $f_{\log}$ are normalized, strictly increasing, and submodular; diminishing returns need not be strict.*

Proof. For $S \subseteq T$ and $e \notin T$, $A_S \preceq A_T$ implies $(A_S + tI)^{-1} \succeq (A_T + tI)^{-1}$, hence $r_e(t;S) \ge r_e(t;T) > 0$. Since $\log(1 + \alpha x)$ increases with $x > 0$ and the integration weight in (11) is positive, integration gives $\Delta_{f_s}(e \mid S) \ge \Delta_{f_s}(e \mid T) > 0$. Equality can occur for candidates in separate connected components. The matrix determinant lemma similarly yields

$$\Delta_{f_{\log}}(e \mid S) = \log(1 + \alpha b_e^\top A_S^{-1} b_e), \tag{12}$$

so the same argument applies to $f_{\log}$. Normalization holds by definition. □

At each fixed shift, the logarithmic term in (11) is the marginal of a shifted log-determinant gain, itself monotone and submodular. Section IV builds a finite surrogate from such terms with a controlled approximation error.

Negative exponents: The inverse trace $J_s(S) = \mathrm{Tr}(A_S^{-s})$, $0 < s < 1$, a regularized analogue of resistance and coherence objectives [1], [2], strictly decreases with edge additions by (7) and (9), but it is not supermodular in general: a companion submission gives an exact four-vertex counterexample [12], and a related network-coherence claim was likewise corrected [13]. For positive powers, we next discretize (11) while preserving its diminishing returns.

IV. Algorithms

The positive integral of Lemma 1 is not yet an efficient marginal oracle: it must be discretized, evaluated reliably, and reused across greedy rounds and related queries. FastFrac combines a structure-preserving surrogate with an offline scalar index and two online execution paths (Fig. 2).

A. Structure-Preserving Gauss–Jacobi Quadrature

Although (11) expresses each marginal through shifted resistances, approximating the integral independently across candidates and rounds need not define a consistent submodular objective. FastFrac instead fixes one positive-weight Gauss–Jacobi rule throughout selection, so all approximate marginals belong to the same surrogate function. For a maximum index-supported budget $K \ge k$, choose a spectral bound

$$I \preceq A_S + u\alpha b_e b_e^\top \preceq \Lambda I, \qquad |S| < K, \quad 0 \le u \le 1. \tag{13}$$

The choice $\Lambda = 1 + 2\alpha(d_{\max}(L_0) + K) > 1$ is valid, where $d_{\max}$ is the maximum weighted degree, since $\lambda_{\max}(L_0) \le 2d_{\max}$ and $\|b_e b_e^\top\| = 2$. Set $\mu_q = \sqrt{\Lambda}$ and $w_s(x) = (1-x)^{-s}(1+x)^{s-1}$. Since both endpoint exponents of w_s exceed -1, its q-point Gauss–Jacobi rule (x_j, ω_j) has $x_j \in (-1,1)$, $\omega_j > 0$, and degree-$2q-1$ exactness. Define

$$t_j = \mu_q \frac{1 + x_j}{1 - x_j}, \qquad W_j = \frac{2 s c_s \mu_q^s \omega_j}{1 - x_j} > 0, \tag{14}$$

and use these nodes and weights throughout selection. The surrogate is

$$g_q(S) = \sum_{j=1}^{q} W_j [\log\det(A_S + t_j I) - \log\det(A_0 + t_j I)], \tag{15}$$

with marginal $\Delta_{g_q}(e \mid S) = \sum_j W_j \log(1 + \alpha r_e(t_j;S))$. Each marginal now uses only q resistance scalars; no log-determinants need evaluation. Positive weights preserve the submodularity of the shifted log-determinant gains. The next result also controls the surrogate's error.

Theorem 2. *Every finite $q \ge 1$ gives a normalized, strictly increasing, submodular g_q. Under* (13)*, put*

$$\rho = \frac{\Lambda^{1/4} + 1}{\Lambda^{1/4} - 1}, \qquad \eta_q = \frac{4}{(\rho^q + \rho^{-q})^2} \le 4\rho^{-2q}. \tag{16}$$

For $|S| < K$ and $e \notin S$,

$$(1 - \eta_q)\Delta_{f_s}(e \mid S) \le \Delta_{g_q}(e \mid S) \le \Delta_{f_s}(e \mid S). \tag{17}$$

Consequently $(1 - \eta_q) f_s(S) \le g_q(S) \le f_s(S)$ for $|S| \le K$. Relative loss at most η follows whenever

$$q \ge \max\left\{1, \left\lceil \frac{\log(4/\eta)}{2\log\rho} \right\rceil \right\}. \tag{18}$$

Proof. Each shifted log-determinant gain is increasing and submodular by Theorem 1; positive summation proves the first claim. For the error bound, substitute $t = \mu_q(1+x)/(1-x)$ in (11). Since $t^{s-1}dt = 2\mu_q^s(1+x)^{s-1}(1-x)^{-s-1}dx$,

$$\Delta_{f_s} = 2 s c_s \mu_q^s \int_{-1}^{1} w_s(x)\chi(x)\, dx, \quad \chi(x) = \frac{\log(1 + \alpha r_e(t(x);S))}{1 - x}. \tag{19}$$

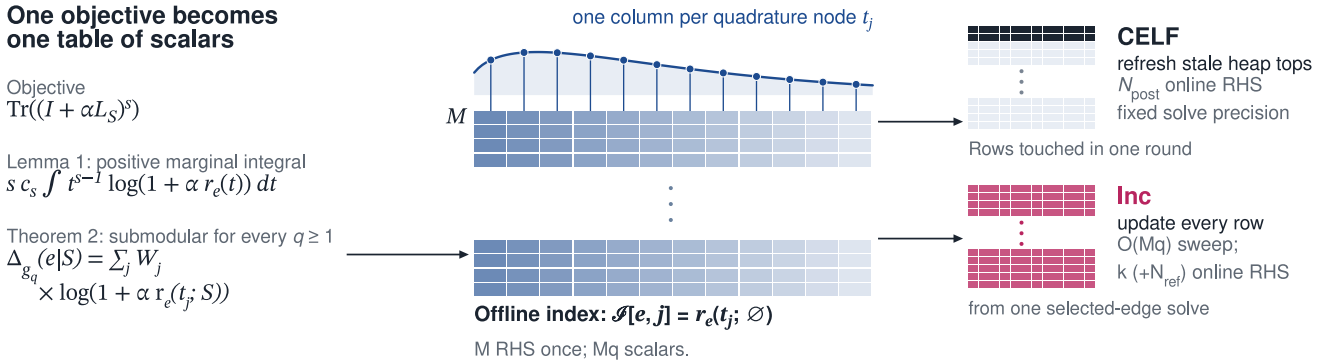


Fig. 2. FastFrac overview. Positive quadrature yields shifted-resistance scalars shared by CELF and Inc. N_{post} counts CELF refreshes after initialization; N_{ref} counts additional Inc refinements. The integrand shape, table shading, and touched rows are schematic.

For $X_u = A_S + u\alpha b_e b_e^\top$, integrating the derivative of $\log\det(X_u + tI)$ gives

$$\chi(x) = \int_0^1 \alpha b_e^\top [(1-x)X_u + \mu_q(1+x)I]^{-1} b_e\, du. \tag{20}$$

Spectral decomposition expresses this as a nonnegative mixture of $a_\lambda(x) = [(\lambda+\mu_q)+(\mu_q-\lambda)x]^{-1}$, with $1 \leq \lambda \leq \Lambda$. Write $\mathrm{Int}[a] = \int_{-1}^1 w_s a$ and $\mathrm{GJ}_q[a] = \sum_j \omega_j a(x_j)$ for integration and quadrature.

The Hermite polynomial P_{Herm} matching a_λ and its first derivative at all q nodes has degree at most $2q-1$, and

$$a_\lambda(x) - P_{\text{Herm}}(x) = \frac{a_\lambda^{(2q)}(\xi_x)}{(2q)!}\prod_j (x-x_j)^2 \geq 0$$

for some $\xi_x \in (-1,1)$, because the even derivative equals $(2q)!(\mu_q-\lambda)^{2q} a_\lambda(x)^{2q+1} \geq 0$. Hence $\mathrm{Int}[a_\lambda] \geq \mathrm{Int}[P_{\text{Herm}}] = \mathrm{GJ}_q[P_{\text{Herm}}] = \mathrm{GJ}_q[a_\lambda]$. If $\lambda = \mu_q$, the rule is exact. Otherwise put $z = -(\lambda+\mu_q)/(\mu_q-\lambda)$, $|z| > 1$, and use the Chebyshev polynomial T_q to define $p(x) = a_\lambda(x)[1 - T_q(x)^2/T_q(z)^2]$. Its numerator vanishes at z, so p is a polynomial of degree at most $2q-1$. Since $|T_q(x)| \leq 1$ for $|x| \leq 1$,

$$(1-T_q(z)^{-2})a_\lambda \leq p \leq a_\lambda,$$
$$\mathrm{GJ}_q[a_\lambda] \geq \mathrm{GJ}_q[p] = \mathrm{Int}[p] \geq (1-T_q(z)^{-2})\,\mathrm{Int}[a_\lambda].$$

The balanced choice $\mu_q = \sqrt{\Lambda}$ ensures $|z| \geq (\sqrt{\Lambda}+1)/(\sqrt{\Lambda}-1) = (\rho+\rho^{-1})/2$. Monotonicity of T_q on $[1,\infty)$ and $T_q((\rho+\rho^{-1})/2) = (\rho^q+\rho^{-q})/2$ give $T_q(z)^{-2} \leq \eta_q$. Integrating these scalar bounds through the positive mixture (20) proves (17). Telescoping along any path to S proves the function bound without an extra $|S|$ factor. Solving $4\rho^{-2q} \leq \eta$ proves (18), and $\log\rho = \Theta(\Lambda^{-1/4})$ gives the order $O(\Lambda^{1/4}\log(1/\eta))$. □

B. Shared-Krylov Multi-Shift CG

Each surrogate marginal requires q shifted systems $(A_S + t_j I)x_j = b_e$. Rather than solve them independently, we exploit their common Krylov space: for $A = A_S$ and $b = b_e$, binomial expansion gives $\mathscr{K}_h(A+t_jI, b) = \mathscr{K}_h(A,b)$, where $\mathscr{K}_h(A,b) = \mathrm{span}\{b, Ab, \ldots, A^{h-1}b\}$. Here h is the number of Lanczos steps, V_h contains the orthonormal Krylov basis, and T_h is the corresponding $h \times h$ tridiagonal projection. The next Lanczos vector and recurrence coefficient are v_{h+1} and β_{h+1}, respectively, while e_1 and e_h denote standard basis vectors in $\mathbb{R}^h$. Lanczos gives $AV_h = V_hT_h + \beta_{h+1}v_{h+1}e_h^\top$ and $b = \|b\|V_he_1$. The shifted Galerkin quantities are

$$\begin{aligned} y_j &= \|b\|V_h(T_h+t_jI)^{-1}e_1, \\ d_j^{\text{CG}} &= \|b\|^2 e_1^\top(T_h+t_jI)^{-1}e_1, \\ \|z_j\| &= \|b\|\beta_{h+1}\left|e_h^\top(T_h+t_jI)^{-1}e_1\right|, \quad z_j = b-(A+t_jI)y_j. \end{aligned} \tag{21}$$

The q shifts share h sparse matrix–vector products (SpMVs) plus $O(qh)$ scalar work [14]. CELF needs only scalar short recurrences; Inc reconstructs selected-edge vectors at $O(nqh)$ cost and $O(nq)$ workspace using shifted directions. One shared multi-shift solve counts as one RHS, *not* one SpMV.

Shared Krylov iterations reduce the cost of evaluating shifted systems, but efficient lazy selection also requires reliable comparisons. A cached marginal estimate cannot safely prune other candidates unless it remains a valid upper bound. We therefore use solver residuals to bound each shifted resistance from below and above, then transfer these intervals to the surrogate marginal through the increasing logarithmic function.

Lemma 2. *Let $B_j = A + t_jI$, $d_j = b^\top B_j^{-1}b$, $z_j = b - B_jy_j$, and $d_j^{\text{var}} = 2b^\top y_j - y_j^\top B_j y_j$. Then*

$$\begin{aligned} d_j - d_j^{\text{var}} &= z_j^\top B_j^{-1} z_j, \\ d_j &\in [\max(0, d_j^{\text{var}}),\ d_j^{\text{var}} + \|z_j\|^2/(1+t_j)]. \end{aligned} \tag{22}$$

Let $t_{\min} = \min_j t_j$, $\kappa = (\Lambda+t_{\min})/(1+t_{\min})$ and $\gamma = (\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$. For $0 < \theta < 1$ and $\kappa > 1$, zero-start CG with

$$h \geq \left\lceil \frac{\log(4\kappa/\theta)}{2\log(1/\gamma)} \right\rceil \tag{23}$$

gives marginal bounds $L_e \leq \Delta_{g_q}(e \mid S) \leq U_e$ with $L_e \geq (1-\theta)U_e$ in exact arithmetic.

Proof. Expand the squared B_j-norm of $B_j^{-1}b - y_j$ and use $B_j \succeq (1+t_j)I$ to obtain (22). Galerkin orthogonality gives $y_j^\top z_j = 0$,

hence $d_j^{\mathrm{var}} = d_j^{\mathrm{CG}} \geq 0$. The CG energy-error bound [15] is $E_j = d_j - d_j^{\mathrm{CG}} \leq \varepsilon_{\mathrm{CG}} d_j$, $\varepsilon_{\mathrm{CG}} = 4\gamma^{2h}$. Since $\|z_j\|^2 \leq (\Lambda + t_j)E_j$, the interval endpoints obey $d_j^- \geq (1-\varepsilon_{\mathrm{CG}})d_j$ and $d_j^+ \leq [1 + (\kappa-1)\varepsilon_{\mathrm{CG}}]d_j$. For $\phi(d) = \log(1+\alpha d)$, concavity and $\phi(0) = 0$ give $\phi(cd) \geq c\phi(d)$ for $c \leq 1$ and the reverse inequality for $c \geq 1$. Therefore, with

$$L_e = \sum_j W_j \phi(d_j^-), \qquad U_e = \sum_j W_j \phi(d_j^+), \tag{24}$$

we have $U_e - L_e \leq \kappa\varepsilon_{\mathrm{CG}} \Delta_{g_q}(e \mid S) \leq \kappa\varepsilon_{\mathrm{CG}} U_e$. Equation (23) ensures $\kappa\varepsilon_{\mathrm{CG}} \leq \theta$. □

C. Offline Scalar Index and Query Model

The fixed quadrature shifts allow initial resistance values to be computed once and reused. Let $\mathscr{C}$ denote the candidate pool C introduced in Section II, with $M = |\mathscr{C}|$. For a fixed base graph G_0, parameters α, s, quadrature order q, spectral bound Λ, and maximum supported budget K, define

$$\mathscr{I}[e, j] = r_e(t_j; \varnothing), \qquad e \in \mathscr{C}, \quad j = 1, \ldots, q. \tag{25}$$

Building the index takes M shared solves on A_0 and stores Mq scalars, without an $M \times M$ cross matrix or persistent solution vectors.

A query over $\mathscr{C}' \subseteq \mathscr{C}$ with $k \leq \min(K, |\mathscr{C}'|)$ copies only the $|\mathscr{C}'|q$ relevant entries and starts from A_0. Each query updates its own copy, leaving the shared index unchanged. Since the spectral bound and numerical error budgets were chosen for K, they also apply to these restricted queries. For prescribed-error Inc, the initialization tolerance must satisfy $\varepsilon_0 \leq \nu/[2(1 + 2\alpha K/(1 + t_{\min}))]$, where ν is the target relative cache-error tolerance defined in Section IV-E.

Changing s, q, or the quadrature rule changes the indexed shifts or weights, while changing α or G_0 changes the shifted systems. These changes, or increasing the supported budget beyond K, generally require rebuilding the index. Candidates outside $\mathscr{C}$ require additional indexed rows. Reuse therefore avoids repeated initialization solves, but not the online selection work.

D. FastFrac-CELF: Certified Lazy Selection

By submodularity, an earlier marginal upper-bounds its value after more edges are selected. FastFrac-CELF initializes a max heap from (25), with fixed edge-ID ties and evaluation timestamps. It refreshes stale leading keys only as needed, avoiding full-pool solves when most upper bounds are already below the leader.

With finite solves, heap keys use the *upper endpoints* from Lemma 2, including rounding allowances; refreshing retains the smaller valid upper bound. Overlapping leaders are refined until

$$L_e \geq (1-\theta) \max_{f \notin S} U_f \tag{26}$$

holds. For $\theta > 0$, this condition certifies a $(1-\theta)$-approximate maximizer of the current surrogate marginal; it does not necessarily identify the exact greedy maximizer. Equal upper bounds enter refinement to avoid cycling; reinsertions without it require a strictly smaller upper endpoint. Lemma 2 guarantees termination for $\theta > 0$, including exact ties.

Algorithm 1 FastFrac-Inc: scalar cache and selected-edge reconstruction

1) Copy $\widehat{r}_{e,j} \leftarrow \mathscr{I}[e, j]$ for $e \in \mathscr{C}'$, $j \leq q$; set $S \leftarrow \varnothing$, $A \leftarrow A_0$.
2) Repeat k times:
3) Select unselected i maximizing $\sum_j W_j \log(1 + \alpha \widehat{r}_{i,j})$, with fixed ID tie priority.
4) Solve $(A + t_j I)v_i^{(j)} = b_i$ for all j with one Multi-Shift CG call, and set $d_i^{(j)} \leftarrow b_i^\top v_i^{(j)}$ from these solutions rather than from the cache.
5) For every unselected $e = (u, v) \in \mathscr{C}'$ and every j: $\widehat{r}_{e,j} \leftarrow \widehat{r}_{e,j} - \alpha[v_i^{(j)}(u) - v_i^{(j)}(v)]^2/(1 + \alpha d_i^{(j)})$.
6) $S \leftarrow S \cup \{i\}$, $A \leftarrow A + \alpha b_i b_i^\top$.

Proposition 1. *With exact oracle values, zero acceptance slack ($\theta = 0$), and the same total tie priority,* FastFrac-CELF *and exhaustive greedy on g_q return the same ordered edge sequence.*

Proof. By Theorem 2, a key evaluated at an earlier $S_e \subseteq S$ upper-bounds the current marginal. Once the maximum key has been refreshed at S, it attains the largest current gain. Any higher-priority candidate with the same marginal has a key at least as large and would have been popped first. Both algorithms choose the same first edge, and induction on their common prefix proves every later choice. □

E. FastFrac-Inc: Rank-One Resistance Maintenance

When many candidates have similar gains, CELF may refresh many stale keys. FastFrac-Inc instead updates all cached resistances after each selection. For the selected edge i, one shared-Krylov solve reconstructs $v_i^{(j)} = (A_S + t_j I)^{-1} b_i$ and $d_i^{(j)} = b_i^\top v_i^{(j)}$. Multiplying (9) by $b_e^\top$ and b_e gives, for $e = (u, v)$,

$$r_e(t_j; S \cup \{i\}) = r_e(t_j; S) - \frac{\alpha[v_i^{(j)}(u) - v_i^{(j)}(v)]^2}{1 + \alpha d_i^{(j)}}. \tag{27}$$

Only the two endpoint entries of each selected-edge solution are needed per candidate. For a query over $\mathscr{C}'$, all active cache rows can therefore be updated in $O(|\mathscr{C}'|q)$ scalar work per round, without an $M \times M$ cross matrix. The selected-edge solution vectors are temporary.

With exact initialization and solves, induction on (27) proves that the cache stays exact. Algorithm 1 thus returns surrogate greedy with one online RHS per selected edge. Figure 3 contrasts CELF refreshes with Inc's scalar sweep on SBM-400.

Cache errors and residual intervals. With inexact solves, initialization and rank-one updates introduce cache error. Prescribed-error mode budgets residuals in advance; adaptive mode propagates intervals and refines ambiguous comparisons. Let S_r be the selected set after r rounds, $r_{e,j}^{(r)} = r_e(t_j; S_r)$, and $\widehat{r}_{e,j}^{(r)}$ its cached value. If every correction uses exact current solutions and fresh denominators, an initial error stays fixed in absolute value:

$$\widehat{r}_{e,j}^{(r)} - r_{e,j}^{(r)} = \widehat{r}_{e,j}^{(0)} - r_{e,j}^{(0)}. \tag{28}$$

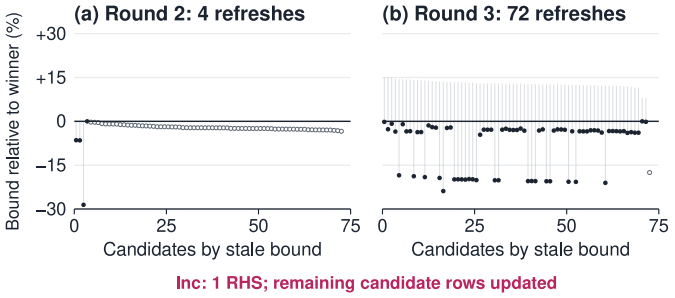


Fig. 3. Stale bounds and CELF refreshes on SBM-400 ($s = 0.10$, $\alpha = 50$). Each panel shows the top 73 remaining candidates ranked by stale bound, with values relative to the round's winner. Lines connect stale bounds to refreshed values; open circles indicate candidates not refreshed. Values come from an exact-objective replay reproducing the recorded selections and RHS counts.

Let $a_j = 1+t_j$, $\upsilon_j = 1+2\alpha k/a_j$. Since $A_{S_r}+t_jI \preceq \upsilon_j(A_0+t_jI)$, inverse order implies $r_{e,j}^{(r)} \geq r_{e,j}^{(0)}/\upsilon_j$; relative initialization error ε_0 can therefore grow to $\upsilon_j\varepsilon_0$.

For approximate selected solution y_j, put $z_j = b_i - (A_S + t_jI)y_j$, $\widehat{c}_{e,j} = y_j(u) - y_j(v)$, and $\varepsilon_{\times,j} = \sqrt{2}\|z_j\|/(1+t_j)$. Cauchy–Schwarz gives $|b_e^\top (A_S+t_jI)^{-1}b_i - \widehat{c}_{e,j}| \leq \varepsilon_{\times,j}$. If $d_i \in [d_i^-, d_i^+]$ comes from (22), the exact subtracted correction $c_{e,j}^{\mathrm{corr}}$ satisfies

$$\frac{\alpha \max(|\widehat{c}_{e,j}| - \varepsilon_{\times,j}, 0)^2}{1+\alpha d_i^+} \leq c_{e,j}^{\mathrm{corr}} \leq \frac{\alpha(|\widehat{c}_{e,j}| + \varepsilon_{\times,j})^2}{1+\alpha d_i^-}. \tag{29}$$

Subtract its upper endpoint from the old lower bound and its lower endpoint from the old upper bound, clipping at zero and optionally at $[2/(\Lambda+t_j), 2/(1+t_j)]$. With rounding allowances added, these are the current marginal intervals that adaptive comparisons use.

Lemma 3. *Let $a = 1+t_{\min}$, $\upsilon = 1+2\alpha k/a$, and $0 < \nu < 1$. Suppose initialization has relative error $\varepsilon_0 \leq \nu/(2\upsilon)$. Use the fresh denominator $\widehat{d}_i = \max(0, d_i^{\mathrm{var}})$ at every update. If selected shifted solves have relative residual at most*

$$\zeta \leq \min\left\{1, \frac{a}{2\alpha}, \frac{\nu a^2}{16\alpha k(\Lambda + t_{\min})}\right\}, \tag{30}$$

all cached resistances have relative error at most ν through k updates, in exact arithmetic with inexact solves.

Proof. For one shift, write $c = b_e^\top (A_S+t_jI)^{-1}b_i$ and $\widehat{c}$ for its computed value. Then $\|b_e\| = \|b_i\| = \sqrt{2}$ gives $|\widehat{c}-c| \leq 2\zeta/a_j$, $|c| \leq 2/a_j$, and $0 \leq d_i - \widehat{d}_i \leq 2\zeta^2/a_j$. Both denominators are at least one, so

$$\left|\frac{\alpha\widehat{c}^2}{1+\alpha\widehat{d}_i} - \frac{\alpha c^2}{1+\alpha d_i}\right| \leq \alpha|\widehat{c}^2 - c^2| + \alpha^2 c^2|\widehat{d}_i - d_i|$$
$$\leq \frac{8\alpha\zeta + 4\alpha\zeta^2}{a_j^2} + \frac{8\alpha^2\zeta^2}{a_j^3} \leq \frac{16\alpha\zeta}{a_j^2}.$$

The last step uses $\zeta \leq 1$ and $\alpha\zeta/a_j \leq 1/2$. Summing at most k errors and dividing by $r_{e,j}^{(r)} \geq 2/(\Lambda+t_j)$ bounds relative correction error by $8\alpha k\zeta(\Lambda+t_j)/(1+t_j)^2$. The function $(1+t)^2/(\Lambda+t)$ increases for $t \geq 0$, as its derivative has numerator $(1+t)(2\Lambda+t-1) > 0$. Thus (30) makes this contribution at most $\nu/2$. The initial error contributes at most $\upsilon\varepsilon_0 \leq \nu/2$ by (28), completing the bound. □

Solve budgets for the two modes. The CG estimate $\|z_h\|/\|b_i\| \leq 2\sqrt{\kappa}\gamma^h$ makes the following selected-vector budget sufficient:

$$h \geq \left\lceil \frac{\log(2\sqrt{\kappa}/\zeta)}{\log(1/\gamma)} \right\rceil. \tag{31}$$

For initialization, define κ_0, γ_0 using a bound $\Lambda_0 \geq \lambda_{\max}(A_0)$; $h_0 \geq \lceil \log(4/\varepsilon_0)/(2\log(1/\gamma_0)) \rceil$ initialization iterations suffice. With these budgets, prescribed-error Inc uses exactly k online RHS. Adaptive Inc propagates (29) and refines overlapping heap leaders until (26) holds, using $k + N_{\mathrm{ref}}$ online RHS for N_{ref} data-dependent refinements.

F. End-to-End Guarantee and Complexity

Combining the one-sided quadrature bound with each path's certified selection quality gives a guarantee for the original fractional-trace objective.

Theorem 3. *Suppose* (17) *holds with $\eta_q \leq \eta < 1$ and the edge e_r selected at each state S_r has surrogate marginal at least $(1-\theta)\max_e \Delta_{g_q}(e \mid S_r)$, where $0 \leq \theta < 1$. For $c = (1-\eta)(1-\theta)$,*

$$f_s(S_k) \geq [1-(1-c/k)^k]\,\mathrm{OPT} \geq (1-e^{-c})\,\mathrm{OPT}. \tag{32}$$

For $0 < \varepsilon < 1$, choosing $\eta = \theta = \varepsilon/2$ gives the ratio $1-1/e-\varepsilon$. Sufficient CELF parameters are

$$q \geq \left\lceil \frac{\log(8/\varepsilon)}{2\log\rho} \right\rceil, \qquad h \geq \left\lceil \frac{\log(8\kappa/\varepsilon)}{2\log(1/\gamma)} \right\rceil. \tag{33}$$

Inc has the same guarantee with $\eta_q \leq \varepsilon/2$, $\nu = \varepsilon/4$, and the budgets of Lemma 3.

Proof. The one-sided quadrature and selection bounds imply $\Delta_{f_s}(e_r \mid S_r) \geq c\max_e \Delta_{f_s}(e \mid S_r)$. For an optimal set O, monotonicity and submodularity give

$$\mathrm{OPT} - f_s(S_r) \leq f_s(S_r \cup O) - f_s(S_r)$$
$$\leq \sum_{e \in O \setminus S_r} \Delta_{f_s}(e \mid S_r) \leq k\max_e \Delta_{f_s}(e \mid S_r).$$

The gap therefore contracts by at most $1-c/k$ per round; starting from OPT and applying $1-x \leq e^{-x}$ proves (32). For $\eta = \theta = \varepsilon/2$, $1-c \leq \varepsilon$ and $e^{-c} - e^{-1} = \int_c^1 e^{-u}\,du \leq 1-c$. Equations (18) and (23) give (33); (26) supplies the selection bound.

For Inc, concavity of $\phi(d) = \log(1+\alpha d)$ transfers relative resistance error ν into $(1-\nu)\phi(d) \leq \phi(\widehat{d}) \leq (1+\nu)\phi(d)$. Positive summation and maximization of approximate scores therefore give selection factor $(1-\nu)/(1+\nu)$. Apply the same gap recurrence with $c = (1-\eta_q)(1-\nu)/(1+\nu)$ and observe $1-c \leq \eta_q + 2\nu/(1+\nu) \leq \varepsilon$ under the stated choices. Lemma 3 supplies the resistance bound. □

TABLE I
FIRST-QUERY COST: OFFLINE INDEX PLUS ONLINE SELECTION (FASTFRAC SHADED)

Method	RHS (off.+on.)	Arithmetic	Storage
Dense greedy	N_k+1 eig.	$O(N_k n^3)$	$O(n^2+M)$
Naive quadrature greedy	N_k	$O(N_k C_{\mathrm{sc}})$	$O(n+Mq)$
CELF	$M+N_{\mathrm{post}}$	$O(MC_{\mathrm{sc},0}+N_{\mathrm{post}}C_{\mathrm{sc}})$ $+O(N_{\mathrm{lazy}}\log M)$	$O(n+Mq)$
Inc, prescribed	$M+k$	$O(MC_{\mathrm{sc},0}+k(C_{\mathrm{vec}}+Mq))$	$O(Mq+nq)$
Inc, adaptive	$M+k+N_{\mathrm{ref}}$	$O(MC_{\mathrm{sc},0}+k(C_{\mathrm{vec}}+Mq)$ $+N_{\mathrm{ref}}(C_{\mathrm{vec}}+\log M))$	$O(Mq+nq)$

$N_k = kM-k(k-1)/2$ counts exhaustive-greedy evaluations; N_{lazy} counts CELF evaluations including initialization and $N_{\mathrm{post}} = N_{\mathrm{lazy}} - M$ its refreshes at fixed precision. C_{sc}, $C_{\mathrm{sc},0}$, and C_{vec} are current scalar, initial scalar, and selected-vector solve costs. N_{ref} counts adaptive Inc refinements; further CELF precision refinements incur separate RHS costs. Reused queries omit M initial RHS and copy $O(|\mathscr{C}'|q)$ entries. Sparse graph storage is excluded.

For the cost analysis, let $D = O(n+m+k)$ denote one SpMV and D_0 its base-graph cost. Scalar, selected-vector, and index-initialization solves cost

$$C_{\mathrm{sc}} = hD+qh, \quad C_{\mathrm{vec}} = hD+nqh+qD, \quad C_{\mathrm{sc},0} = h_0D_0+qh_0,$$

Here qD covers explicit residual checks. Naive greedy uses $N_k = kM-k(k-1)/2$ evaluations. CELF uses $M \le N_{\mathrm{lazy}} \le N_k$. This bound counts at most one candidate evaluation per round at a fixed solve precision; additional precision refinements, when required, incur separate solve costs. There are $N_{\mathrm{post}} = N_{\mathrm{lazy}} - M$ online refreshes. Index construction takes M RHS once; a reused query reads $O(Mq)$ entries and uses N_{post}, k, or $k+N_{\mathrm{ref}}$ online RHS for CELF, prescribed Inc, or adaptive Inc, respectively. Table I reports first-query costs; vector workspace is $O(nq)$ and quadrature-rule construction $O(q^2)$.

V. EXPERIMENTAL EVALUATION

We examine two questions motivated by the preceding analysis: whether the fractional exponent changes selected edges and their diffusion effects, and whether FastFrac can optimize this objective efficiently at scale. We first test spectral tunability, then assess solution quality, query cost, large-network scalability, the two online paths, and numerical accuracy.

A. Experimental Settings

Workloads. Table II summarizes the synthetic and SNAP networks. Small-BA, Medium-WS, and Large-PowerCluster use preferential attachment, small-world rewiring, and power-law clustering [16]–[18]; their candidate pools are $1/2$ uniform, $1/4$ two-hop, and $1/4$ degree-stratified inter-cluster pairs. The BA sweep uses $\mathrm{BA}(n,3)$ graphs. SNAP graphs [19] retain all endpoint vertices and are made undirected, with loops and duplicate edges removed. All methods use the same graph and ordered candidate pool.

Two multiscale families isolate spectral preferences and competition among bridge candidates. SBM-400 comprises four 100-vertex blocks with within-block probability 0.30 and two weight-0.02 inter-block bridges, leaving two components [20]. The heterogeneous graph joins four $\mathrm{BA}(100,3)$ blocks with weight scales $(0.2,0.7,2,6)$, lognormal multipliers of log-SD 0.35, and bridge weights $(0.002,0.02,0.2)$. Each pool has 75 inter-cluster bridge and 75 intra-cluster core candidates. SBM bridge endpoints come from the lowest degree quartile and core endpoints from the highest; the heterogeneous bridge pool draws from both weighted-degree quartiles. Ten further graphs from each generator test reproducibility of the spectral trends.

TABLE II
WORKLOADS

Dataset	n	m	M	k
Small-BA	400	1,191	200	10
Medium-WS	2,500	10,000	1,000	20
Large-PowerCluster	10,000	39,976	3,000	30
BA sweep	1k–100k	$\approx 3n$	2,000	20
SBM-400	400	5,990	150	15
Heterogeneous	400	1,167	150	15
SBM-4k	4,000	59,772	600	20
email-Eu-core	1,005	16,064	2,000	20
ca-GrQc	5,242	14,484	2,000	20
ca-CondMat	23,133	93,439	2,000	20
com-DBLP	317,080	1,049,866	2,000	20
com-Amazon	334,863	925,872	2,000	20
web-Google	875,713	4,322,051	2,000	20
roadNet-PA	1,088,092	1,541,898	2,000	20
com-Youtube	1,134,890	2,987,624	2,000	20
as-Skitter	1,696,415	11,095,298	2,000	20
roadNet-CA	1,965,206	2,766,607	2,000	20
com-LiveJournal	3,997,962	34,681,189	2,000	20

Baselines and metrics. Naive evaluates every candidate in every round on the same quadrature surrogate as FastFrac; it isolates the savings from query execution. Dynamic-Logdet (Logdet) applies the same lazy-selection machinery to $f_{\log}$ and tests the value of varying spectral emphasis. Static-EffRes ranks $b_e^\top A_0^{-1} b_e$ once in descending order; Degree-Deficit ranks base endpoint-degree sums in ascending order (weighted where applicable); Random averages ten fixed-seed, budget-feasible uniform subsets. We independently score the fractional objective: small and multiscale graphs use dense spectra, while Medium/Large graphs use high-order path quadrature with checked residuals. Numerical-oracle comparisons additionally use the unintegrated marginal and common-random-number stochastic Lanczos quadrature (CRN-SLQ).

Protocol. Defaults are $s = 0.5$, $\alpha = 1$, $q = 12$, and relative residual tolerance 10^{-7}; uncertain comparisons refine to 10^{-12} with a numerical tie threshold $2\times 10^{-11}\max(1,\mathrm{marginal})$. All timing comparisons use C++17 kernels compiled with `-O3 -march=native -fopenmp`, one BLAS thread, and Python only for orchestration. Main experiments use an AMD EPYC 9V74 (at most eight OpenMP threads); the candidate/budget sweeps and ten-graph replications use an Intel Core Ultra X7 358H (same thread limit); Table VI uses an Intel Xeon Platinum 8481C (18 threads). Methods in each comparison run on the same machine. Unless noted otherwise, timings are means of five runs with fresh initialization and sample standard deviations. They exclude graph generation, compilation, and independent accuracy evaluation. Index construction ends when all initial scalar entries are computed; online time includes copying, scoring, heap operations, and updates. The package includes code, raw timings, selections, solver counters, seeds,

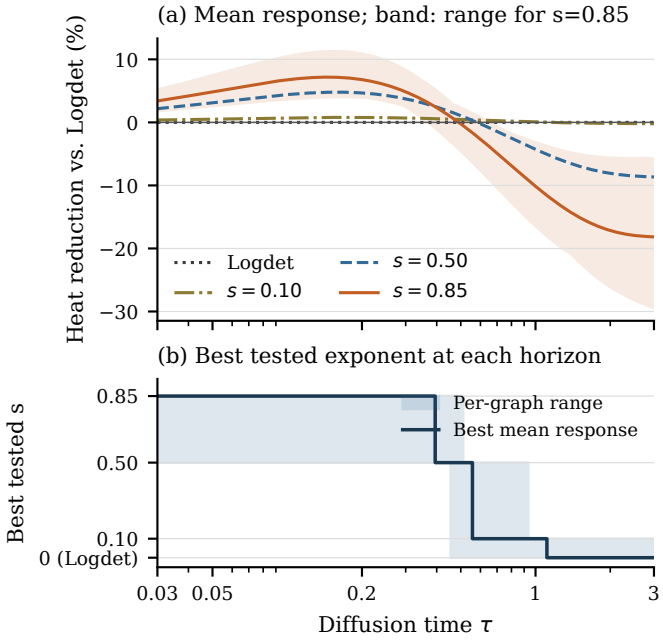


Fig. 4. Post-hoc horizon sensitivity on ten heterogeneous graphs ($\alpha = 50$, $k = 15$). (a) Per-graph $R_s(\tau)$ averaged across graphs; the band shows the range for $s = 0.85$. (b) Best mean response among Logdet and $s \in \{0.10, 0.50, 0.85\}$; shading spans per-graph best settings. Here $s = 0$ denotes Logdet; ties within 10^{-8} percentage points favor smaller s. This discrete comparison is not a continuous optimum or a validated tuning rule.

TABLE III

HEAT-TRACE RESPONSE VERSUS LOGDET: (A) TEN-GRAPH REPLICATIONS; (B) SINGLE-INSTANCE REAL GRAPHS.

s	Swap	Gain	$\tau = 0.05$		$\tau = 0.2$		$\tau = 1$	
SBM								
0.10	13%	+0.06	+0.77 7/10	±0.99	+1.56 7/10	±2.29	−0.42 0/10	±0.53
0.50	31%	+0.56	**+2.00** 10/10	±1.10	**+3.98** 10/10	±2.28	**−1.67** 0/10	±0.76
0.85	35%	+0.27	**+2.27** 10/10	±1.12	**+4.38** 10/10	±2.32	**−2.15** 0/10	±0.78
Heterogeneous								
0.10	6%	+0.08	+0.53 4/10	±0.71	+0.79 4/10	±1.16	+0.06 2/10	±0.29
0.50	18%	+0.84	**+3.12** 10/10	±1.64	**+4.67** 10/10	±2.30	**−4.28** 0/10	±2.83
0.85	27%	+1.12	**+4.86** 10/10	±1.58	**+6.79** 10/10	±2.38	**−10.14** 0/10	±4.51

(a) Heat-trace cells: mean ± SD of $R_s(\tau)$ over ten graphs; orange: FastFrac better, blue: Logdet better, gray: $|\text{mean}| < \text{SD}$. Counts $x/10$: number of graphs where FastFrac is strictly better (ties occur only at $s = 0.10$). Swap: Logdet edges replaced. Gain: objective gain over Logdet (%). $\alpha = 50$, $k = 15$, $q = 12$; Intel machine.

(b) Real graphs: fixed-pool single instances ($\alpha = 50$, $M = 2000$, $k = 20$, $q = 12$)

Graph	s	Swap	$R_s(.05)$	$R_s(.2)$	$R_s(1)$
email-Eu-core	.10	0%	0	0	0
	.50	0%	0	0	0
	.85	25%	+9.37	+9.20	-12.10
ca-GrQc	.10	5%	+0.24	+0.79	+0.73
	.50	30%	+1.08	+3.65	+3.51
	.85	80%	+2.29	+7.90	+4.75

(b) R_s is in percent; Swap $= 1 - |S_s \cap S_{\log}|/k$, where S_s and $S_{\log}$ are the fractional and Logdet sets. Each query restarts from the same base graph and original candidate pool; heat traces use full Laplacian spectra. No seed averaging.

and environment details.

B. Spectral Tunability

We first test whether changing s alters the recommended edges and their effects across diffusion horizons. We evaluate the selected sets using the standard-Laplacian heat-trace reduction $\Delta\mathscr{H}(\tau) = \mathrm{Tr}(e^{-\tau L_0}) - \mathrm{Tr}(e^{-\tau L_S})$, computed from full spectra at $\alpha = 50$ and $k = 15$. Here τ is the evaluation time, distinct from the selection parameter s; unlike (5), this metric uses L_S rather than the fractional generator Q_s. Let $\Delta\mathscr{H}_s(\tau)$ and $\Delta\mathscr{H}_{\log}(\tau)$ denote reductions for the respective selected sets. We report the relative change

$$R_s(\tau) = 100\left(\frac{\Delta\mathscr{H}_s(\tau)}{\Delta\mathscr{H}_{\log}(\tau)} - 1\right)\%. \tag{34}$$

Positive $R_s(\tau)$ indicates greater heat-trace reduction than Logdet; negative values indicate less.

On the heterogeneous reference graph, $s = 0.85$ yields 5.47% more heat-trace reduction than Logdet at $\tau = 0.2$, but 10.20% less at $\tau = 1$; the relative advantage switches near $\tau = 0.44$. The SBM reference graph follows the same pattern (+5.16% and −2.77%). The shortest horizon can also favor a simpler baseline: Degree-Deficit leads at $\tau = 0.05$ on the heterogeneous reference.

Across ten additional graphs per generator (Table III), both $s = 0.50$ and $s = 0.85$ outperform Logdet at $\tau = 0.05$ and 0.2, but underperform it at $\tau = 1$, on every instance. For $s = 0.85$, the heterogeneous mean changes are $+6.79\% \pm 2.38\%$ at $\tau = 0.2$ and $-10.14\% \pm 4.51\%$ at $\tau = 1$. At $s = 0.10$, selections nearly match Logdet, consistent with the limiting relation (4). Evaluated under the corresponding f_s, the mean gain over the Logdet-selected set is at most 1.12% in these settings, despite the more pronounced horizon-specific differences.

The selected edges also change with s. At $\alpha = 50$, the initial four-lowest-mode share of SBM inter-cluster candidates falls from 85.41% at $s = 0.10$ to 2.84% at $s = 0.85$ (Fig. 1(b)). Seven exponents on the heterogeneous reference yield three distinct sets: $s \le 0.25$ (identical to Logdet), $s = 0.50$, and $s \ge 0.70$. Across these sets, common two-hop neighbors rise from 2.40 to 9.47, while weighted degree falls from 6.64 to 4.06; ten heterogeneous replications show the same trend. All SBM selections remain inter-cluster.

Figure 4 extends the ten-graph selections across $\tau \in [0.03, 3]$. At $\tau = 0.2$, $s = 0.85$ leads on nine graphs and $s = 0.50$ on one; at $\tau = 1$, Logdet leads on eight (including ties) and $s = 0.10$ on two. Table III(b) adds real-graph controls: at $s = 0.85$, email-Eu-core reverses sign, whereas ca-GrQc remains positive at all three horizons. Higher-order checks ($q = 69$ and 50, respectively, $\eta_q < 10^{-8}$) preserve five of six sets; the remaining ca-GrQc $s = 0.50$ set changes one edge but preserves the reported heat responses. At $(s, \alpha) = (0.5, 1)$, the eight large SNAP selections differ from Logdet by at most one edge; this does not characterize the full family. Thus separation also occurs on real graphs at $\alpha = 50$, but neither its size nor a horizon reversal is universal.

TABLE IV
SINGLE-QUERY SWEEP ENDPOINTS ON MEDIUM-WS (DEFAULTS). TOTAL SECONDS INCLUDING FRESH INDEX CONSTRUCTION: MEAN ± SAMPLE SD OVER FIVE COMPLETE RUNS.

Setting	CELF (s)	Naive (s)
$M = 250,\ k = 20$	0.059 ± 0.008	0.324 ± 0.054
$M = 4000,\ k = 20$	0.297 ± 0.024	4.605 ± 0.269
$M = 1000,\ k = 5$	0.066 ± 0.003	0.299 ± 0.042
$M = 1000,\ k = 60$	0.207 ± 0.009	3.569 ± 0.224

C. Solution Quality

To assess optimization quality independently of diffusion behavior, we enumerate all $20\binom{15}{4} = 27{,}300$ feasible sets across twenty connected instances ($n = 25$, $M = 15$, $k = 4$). Both FastFrac paths attain a mean 0.9993 OPT and minimum 0.9902 OPT, above the worst-case guarantee of Theorem 3. Logdet is also competitive on this fractional objective (mean 0.9991 OPT, the same minimum), while Static-EffRes, Degree-Deficit, and Random average 0.9910 OPT, 0.9903 OPT, and 0.9026 OPT, respectively.

D. Query Efficiency and Index Cost

To test lazy RHS savings, Naive, CELF, and Inc optimize the same quadrature surrogate. On Large-PowerCluster (Fig. 5), CELF reduces total time from Naive's 59.7 ± 3.6 s to 2.48 ± 0.15 s, a $24.1\times$ speedup including fresh index construction. Shared multi-shift right-hand sides (RHS) fall from 89,565 to 3,031: 3,000 initial and 31 online solves. Inc uses 3,030 RHS and 2.82 ± 0.31 s.

Figure 6 shows BA scaling. At BA-50k, CELF takes 20.1 ± 1.2 s versus Naive's 345 ± 29 s. At BA-100k, Naive was not completed; CELF takes 74.3 ± 2.9 s (4.93 ± 0.18 s online), slightly above Logdet's 73.7 ± 1.8 s.

Single-query pool and budget sweeps. On Medium-WS ($n = 2500$), we vary nested pools $M \in \{250, 500, 1000, 2000, 4000\}$ at $k = 20$, and budgets $k \in \{5, 10, 20, 40, 60\}$ at $M = 1000$, using the defaults. Across the pool scan, CELF online time rises from 37.6 ± 5.5 to 63.0 ± 9.7 ms ($1.68\times$), versus $14.12\times$ for Naive; CELF's online RHS count rises from 20 to 33. At the budget-scan endpoints, CELF total RHS increase from 1004 to 1093, versus 4990 to 58230 for Naive. Table IV reports total times. Both algorithms agree in order throughout all configurations and repeats.

Multi-query index reuse. We test index reuse (Section IV-C) with three independently seeded 100-query workloads per graph (email-Eu-core and ca-GrQc; seeds 20261109–20261111). Each query uniformly samples 500, 1,000, or 1,500 of the original 2,000 candidates and a budget in $\{5, 10, 20\}$, restarting from $(G_0, \varnothing)$. Both modes use identical FASTFRAC-CELF with $s = 0.5$, $\alpha = 1$, $q = 12$, and common $K = 20$ bounds, nodes, and tolerances. Fresh builds only the filtered index per query; reuse builds the full index once and copies requested rows. Paired execution order alternates on the eight-thread Intel machine; each workload is timed once.

TABLE V
MEASURED CUMULATIVE TIME FOR 100 FILTERED QUERIES. MEAN ± SAMPLE SD ACROSS THREE WORKLOAD SEEDS, EACH TIMED ONCE; SPEEDUP IS THE RATIO OF MEAN TIMES.

Graph	Fresh (s)	Reuse (s)	Speedup
email-Eu-core	13.99 ± 0.17	5.42 ± 0.20	$2.58\times$
ca-GrQc	23.63 ± 0.38	6.61 ± 0.40	$3.57\times$

Timing in Table V includes filtering, row copying, graph reset, scoring, heap operations, selection, and refinement; shared loading, query generation, rule construction, and independent checks are excluded. The index remains in memory. All 600 paired queries return identical ordered selections and computed gains. Reuse costs more on query one but becomes cumulatively faster at query two and stays ahead through query 100 in all six workloads (Fig. 7). Per-seed speedups range from 2.48–2.73× and 3.43–3.82×, respectively. Full-index construction averages 0.170 s and 0.366 s; point-value and two-bound arrays occupy 0.576 MB, excluding graph and solver workspace. These are synthetic random filters on real graphs, not production query traces.

E. Large-Scale SNAP Evaluation

On eight SNAP networks at $\alpha = 1$, CELF completes every run and is fastest on seven (Table VI); Naive times are projected, not completed. The largest, com-LiveJournal (4.0 million vertices, 34.7 million edges), takes 8,562 s (2.4 hours), with 26.7 GB peak memory including graph and solver workspace beyond the Mq-scalar index.

TABLE VI
LARGE SNAP NETWORKS ORDERED BY MAXIMUM DEGREE ($\alpha = 1$, $M = 2000$, $k = 20$, $s = 0.5$, $q = 12$). CELF, INC, AND LOGDET TIMES ARE MEASURED SECONDS; NAIVE TIMES ARE PROJECTED HOURS. ITER. DENOTES LANCZOS ITERATIONS PER RHS. ROUNDED r_Q ASSUMES EXACT SURROGATE GREEDY, NOT DEFAULT-RUN CERTIFICATION. INC WAS NOT RUN ON COM-LIVEJOURNAL.

		Total time			Iter.			
Dataset	$d_{\max}$	CELF	Inc	Logdet	CELF	Logdet	Naive (h)	r_Q
roadNet-PA	9	42.2	54.5	**41.3**	22	22	0.2	.632
roadNet-CA	12	**74.9**	106.0	84.3	22	22	0.3	.632
com-DBLP	343	**86.7**	93.9	90.4	106	111	0.4	.632
com-Amazon	549	**89.0**	104.9	95.1	96	103	0.4	.632
web-Google	6,332	**1,381**	1,483	1,670	380	467	6.2	.616
com-LiveJournal	14,815	**8,562**	—	10,876	399	509	36.7	.594
com-Youtube	28,754	**1,166**	1,340	1,746	418	593	5.1	.566
as-Skitter	35,455	**4,262**	4,597	5,935	762	1094	18.2	.555

CELF needs 418 shifted Lanczos iterations per RHS on com-Youtube versus 593 for Logdet. It is 21–33% faster on the three graphs with $d_{\max} > 10^4$, but 2% slower on roadNet-PA. Inc is slower on all seven measured networks, with identical FastFrac selection sequences.

Table VI reports $r_Q = 1 - \exp(-(1 - \eta_{12}))$ under exact surrogate greedy ($\theta = 0$), using the actual spectral bounds $\Lambda = 1 + 2(d_{\max} + 20)$. Certified per-round quality $1 - \theta$ instead gives $1 - \exp(-(1 - \eta_{12})(1 - \theta))$. Default floating-point residuals and tie checks lack outward-rounded enclosures and

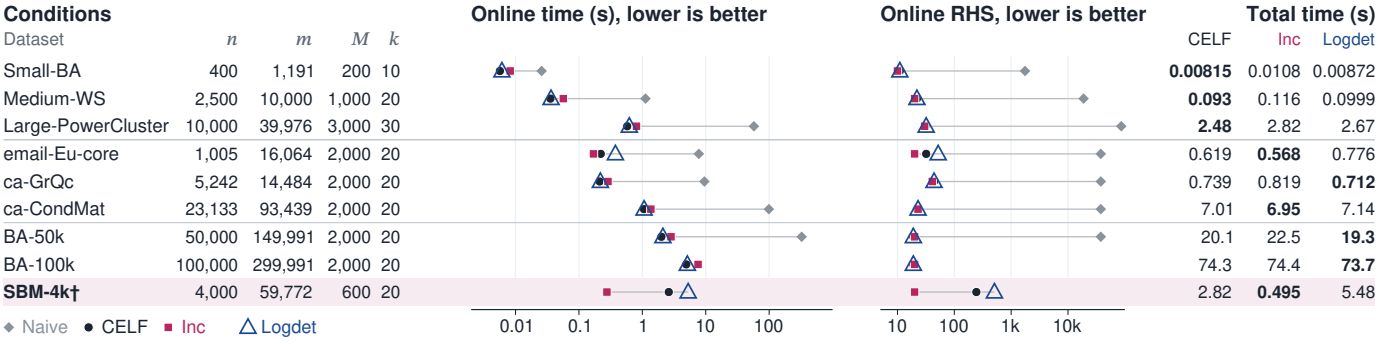


Fig. 5. Query cost per dataset, as means of five runs on the EPYC machine. Plots exclude initialization; the right block includes fresh index construction, with the lowest total in bold. Online standard deviations are below 16% (median 4%). † $s = 0.10, \alpha = 50$; all others $s = 0.5, \alpha = 1$; Naive was not completed on BA-100k or SBM-4k.

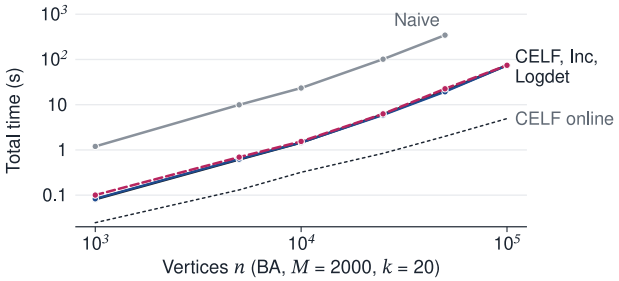


Fig. 6. Total time against graph size on BA graphs ($M = 2000$, $k = 20$, $s = 0.5$, $\alpha = 1$), as means of five runs on the EPYC machine. Naive was not completed at $n = 100000$.

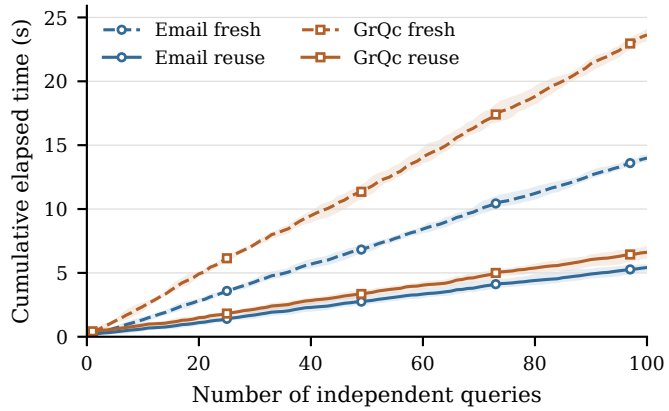


Fig. 7. Cumulative query-processing time on email-Eu-core (Email) and ca-GrQc (GrQc), including one full-index build for reuse or one filtered-index build per fresh query. Lines show means and bands the range over three workload seeds. Every query restarts at the base graph.

per-round θ certificates, so r_Q is not a certified ratio for those outputs. Worst-case iteration bounds scale as $O(\sqrt{\kappa}\log(\kappa/\theta))$ for $0 < \theta < 1$, with $\kappa = (\Lambda + t_{\min})/(1 + t_{\min}) \le \Lambda$; this is not an observed $\sqrt{\Lambda}$ scaling law. The $\alpha = 50$ experiments cover SBM-4k and real ca-GrQc (5,242 vertices), not million-vertex timing workloads.

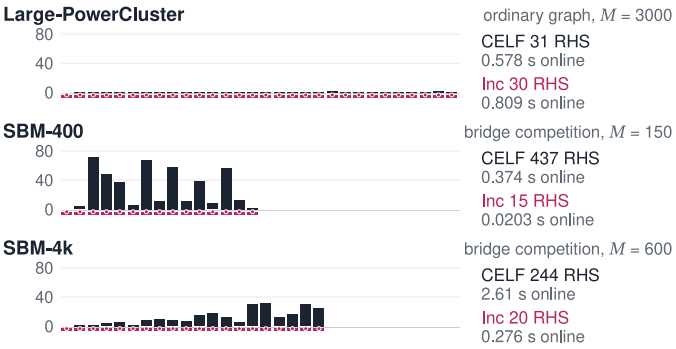


Fig. 8. Online RHS per round: CELF above, Inc below the axis; markers identify small Inc bars. Initialization is excluded. Bars show the first run; times average five online-only EPYC runs.

F. When Inc Outperforms CELF

Near-tied bridge candidates can force repeated CELF refreshes (up to 72 in one SBM round; Figs. 3 and 8). Inc instead uses one selected-edge solve and an $O(Mq)$ scalar sweep per round.

At $s = 0.10$ and $\alpha = 50$ on SBM-400, CELF uses 437 online RHS and takes 0.379 ± 0.027 s in total, while Inc uses 15 online RHS and takes 0.0253 ± 0.00071 s in total, a $15.0\times$ measured total-time speedup. On SBM-4k, Inc achieves a $5.7\times$ total speedup. Across ten additional SBM graphs, Inc is $16.5 \pm 0.9\times$ faster in total at $s = 0.10$; on heterogeneous graphs at $s \ge 0.5$, the paths differ by about 10% or less. Logdet also requires 489 and 508 online RHS on SBM-400 and SBM-4k. These results favor Inc when many bridge candidates compete, rather than as a general replacement for CELF.

G. Numerical Accuracy and Robustness

Figure 9 compares observed relative marginal error with the bound of Theorem 2. On Small-BA, the maximum over 1,955 marginals along the Exact-Eig greedy path falls from 3.47×10^{-4} at $q = 4$ to 1.38×10^{-10} at $q = 12$. For SBM at $\alpha = 50$, errors at $q = 12$ are 1.076×10^{-3} ($s = 0.10$) and 7.749×10^{-4} ($s = 0.50$), below $\eta_{12} = 0.01658$. Every tested $q \ge 8$ recovers the full Exact-Eig sequence for both exponents.

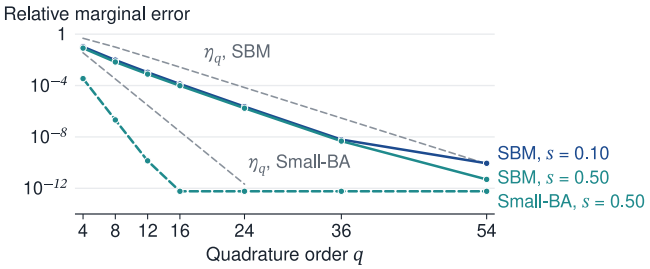


Fig. 9. Quadrature error against the bound η_q of Theorem 2 (gray dashed, independent of s) on Small-BA ($\alpha = 1$) and SBM ($\alpha = 50$). Colored lines give the maximum over all candidates on the Exact-Eig greedy path, and flat tails are rounding.

TABLE VII
ORACLE ROUTES ON MEDIUM-WS, RELATIVE TO FASTFRAC (LOWER IS BETTER; GAIN AND TOP-k: HIGHER)

Route	Time	SpMV	Max. err.	Gain G	Top-k
FastFrac (IBP)	1	1	4.5×10^{-11}	7.257	20/20
Raw-Proj	1.00×	1×	1.1×10^{-9}	7.257	20/20
Raw-FullVec	3.83×	1×	1.1×10^{-9}	7.257	20/20
CRN-SLQ CELF	43.8×	77.3×	0.26	7.000	2/20
CRN-SLQ Greedy	807×	1499×	0.26	7.000	2/20

Means of five instrumented runs (recording overhead included, so times exceed Fig. 5). FastFrac: 0.0928 s, 25,372 SpMVs. SLQ: $p = 128$, $h = 15$. Top-k: set overlap with the deterministic reference.

Across the ten-graph replications, $q = 12$ and $q = 54$ select the same sets in 60/60 cases and the same order in 55/60, with equal final gains. The general guarantee uses the order prescribed by Theorem 2.

Residual refinement. We test whether refinement is needed to reproduce the reference order. On Small-BA at tolerance 10^{-1}, 34 extra solves recover the reference order; without refinement, two positions differ but the set and gain agree. At 10^{-2}, 10^{-4}, and 10^{-7}, both variants match without extra solves. On ca-GrQc, all 21 refinements concern structural ties. Against a tighter 10^{-12} floating-point CELF reference, positional/set agreements are 75%/95%; vertex swaps $3108 \leftrightarrow 4491$, $4347 \leftrightarrow 5079$ map the augmented graphs onto each other, proving identical spectra. Tested gains differ by less than 10^{-12}.

Alternative marginal oracles. Table VII compares complete Medium-WS routes under matched threads. Raw-Proj evaluates (10) via a differentiated shifted LDL recurrence; Raw-FullVec forms explicit shifted vectors. Lazy and exhaustive CRN-SLQ share $p = 128$ fixed Rademacher probes and $h = 15$ Lanczos steps [21]. The deterministic routes agree in selections and gains. IBP matches Raw-Proj in time and SpMVs: its contribution is the positive logarithmic mixture and structure-preserving surrogate, not a standalone speedup.

Sampled estimates are not valid lazy upper bounds: about 37% of CRN-SLQ's cached point keys on Medium-WS fall below later sampled marginals. On Small-BA, the same route has 23% maximum error, 4/10 selected-edge overlap, and 18.1× the runtime of FastFrac. Same-probe lazy and exhaustive SLQ nevertheless give identical orders on both graphs, so no incorrect pruning was observed.

VI. RELATED WORK

A. Spectral Design and Submodular Optimization

Spectral objectives guide network design for effective-resistance minimization [1], network coherence [2], and polarization reduction [3], each emphasizing a particular graph-wide property. Matrix-forest identities [22] connect regularized log-determinants to combinatorial graph structure, while fractional network dynamics [9] motivate varying spectral emphasis.

Classical greedy and CELF exploit submodularity and cached upper bounds [7], [23]. Positive fractional traces are submodular [6]; the negative-power counterexample in POLAR [12] and the coherence correction [13] show why this property must be checked per objective.

	FastFrac	POLAR
Objective	Positive power; submodular	Negative power; non-submodular
Initialization	M scalar RHS	$2d_{\text{sk}}$ sketch RHS
Selection	Residual-controlled lazy/Inc	Proxy top-B, algebraic re-ranking

Here d_{sk} is sketch dimension and B is shortlist size. Sampled point estimates alone do not provide deterministic upper bounds for lazy selection, motivating FastFrac's initial-resistance table. POLAR has conditional ratio–curvature guarantees; its Gaussian certificate does not cover default block-orthogonal sketches. The identity $A_S = \alpha(L_S + \sigma_{\text{POLAR}} I)$ with $\sigma_{\text{POLAR}} = 1/\alpha$ aligns matrix normalization, not objectives.

B. Spectral Graph Queries and Indexing

Specialized algorithms address current-flow group closeness [4], resistance eccentricity [5], resistance diameter [24], and biharmonic distance [25]. For evolving graphs, Liao et al. [26] maintain effective-resistance indexes using random-walk and loop-erased-walk samples. FastFrac instead reuses a fixed base-graph resistance index across filtered queries and updates private state during each query's edge insertions.

C. Matrix-Function Estimation and Numerical Solvers

Hutchinson [27], Hutch++ [28], stochastic Lanczos quadrature [21], and indefinite-matrix estimators [29] reduce trace-estimation cost. Greedy selection additionally requires reliable marginal comparisons; sampled point keys do not certify CELF pruning (Section V-G). Our numerical building blocks include matrix-function representations [8], [10], Lanczos quadrature [15], stability analysis [30], shifted Krylov spaces [14], and Sherman–Morrison updates [11]. SDD solvers, sparsification, and approximate elimination [31]–[33] are complementary primitives.

VII. CONCLUSION

FastFrac combines positive fractional-trace marginals, a reusable resistance index, and lazy or incremental selection under explicit numerical conditions. The tested SLQ and near-tie comparisons show 43.8× and 15.0× speedups; repeated queries demonstrate additional reuse savings. Real and multiscale controls reveal useful but workload-dependent spectral preferences. Future work includes base-graph index maintenance and improved preconditioning.

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